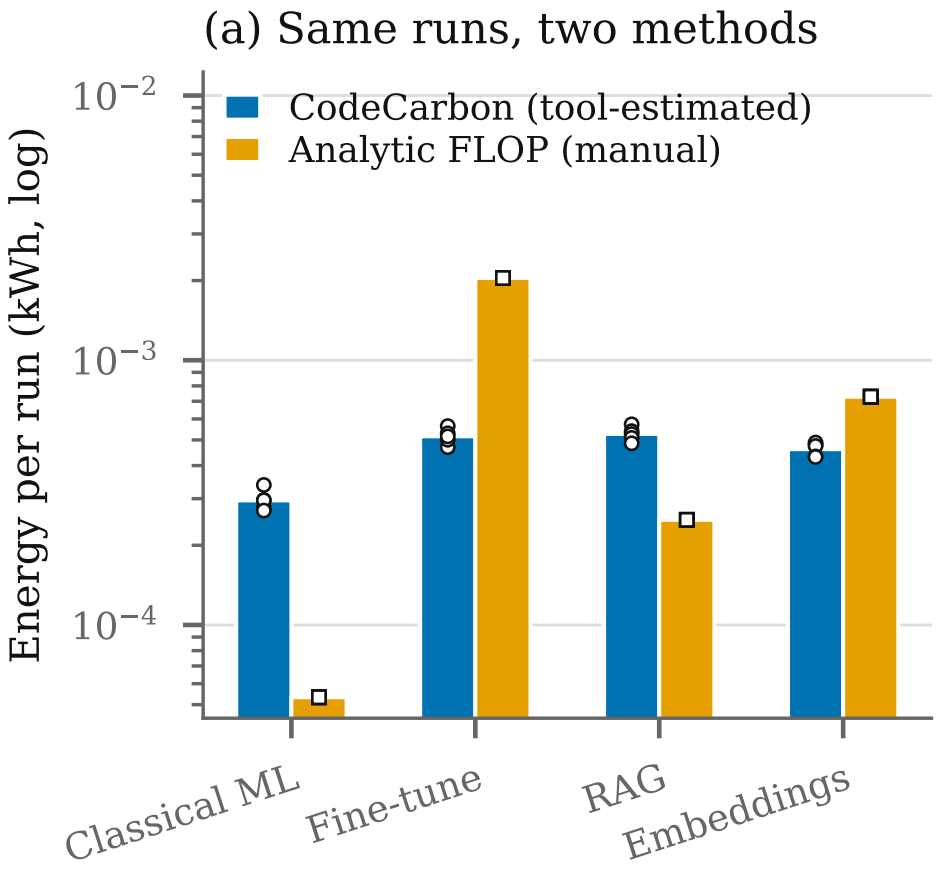
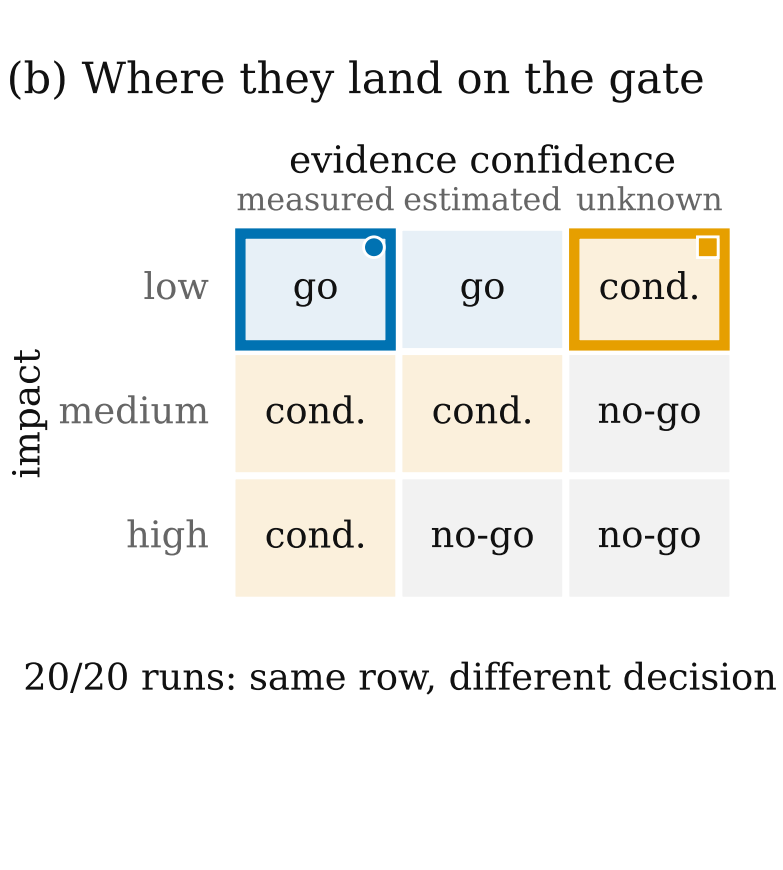

(a) Same runs, two methods
CodeCarbon (tool-estimated)
Analytic FLOP (manual)
Energy per run (kWh, log)
10^{-2}
10^{-3}
10^{-4}
Classical ML
Fine-tune
RAG
Embeddings


(b) Where they land on the gate
evidence confidence
measured
estimated
unknown
impact
low
medium
high
go
go
cond.
cond.
cond.
no-go
cond.
no-go
no-go
20/20 runs: same row, different decision